Cellulose (2019) 26:205–213
https://doi.org/10.1007/s10570-018-2130-7

REVIEW PAPER

Deep eutectic solvents (DESs) for cellulose dissolution: a mini-review

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Received: 11 September 2018 / Accepted: 8 November 2018 / Published online: 17 November 2018
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Abstract Deep eutectic solvents (DESs), which are a novel class of sustainable designer solvents, have attracted considerable attentions in the field of cellulose chemistry. Due to their low cost and analogous physico-chemical properties to ionic liquids, DESs are expected to be alternative solvents for dissolving cellulose. However, at present, the solubility of cellulose in DESs is much lower than in most ionic liquids. In this mini-review, we briefly summarize the current state of knowledge about cellulose dissolution in DESs. By comparing with similar solvents, it was found that the components of current DESs are usually involved in hydrogen bond interaction making difficult their interaction with the hydrogen bond network of cellulose. Accordingly, we propose a strategy that the components which have good hydrogen bond accepting ability, such as Cl^-, OAc^-, $HCOO^-$, $(MeO)_2PO_2^-$, morpholine and imidazole, are promising choices to form DESs for cellulose dissolution. Ultrasound-assisted treatment and adding a surfactant are effective ways to promote cellulose solubility by enhancing the permeability of DESs.

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Keywords Cellulose · Dissolution · Deep eutectic solvents (DESs)

Introduction

Cellulose is the most abundant renewable resource in the world. As a promising material, it offers advantages including edibility, biocompatibility, barrier properties, attractive appearance, non-toxicity, non-polluting and low cost (Imran et al. 2010). The dissolution of cellulose is a critical step for the efficient conversion of this renewable resource into high value-added products, such as regenerated materials (fibers, films, food casings, membranes, sponges, etc.) and cellulose derivatives (esters, ethers, and grafted copolymers) (Liu et al. 2016; Ghasemi et al. 2017). Traditional solvents for cellulose dissolution involve several types of derivatizing solvents and non-derivatizing solvents (Heinze and Koschella 2005). A well-known derivatizing solvent is CS_2 in NaOH used in the industrially important viscose process. In this case, CS_2 reacts with cellulose to form a more easily soluble cellulose xanthogenate. One example of non-derivatizing solvent is *N*-methyl-morpholine-*N*-oxide (NMMO). Lyocell fiber which is spun from NMMO solution is proposed to be an eco-friendly alternative to cotton and other technical fibers. These solvents, however, exhibit some drawbacks. For instance, the viscose process leads to environmental pollution from

the use of CS_2; NMMO has a limited thermal stability which results in a safety risk. With sustainable development becoming top of the agenda, researchers are continuously looking for more efficient, greener, safer and cheaper solvents to dissolve cellulose.

The ionic liquids are considered environmentally friendly or "green" media due to their favorable properties (Zhao 2003). Swatloski et al. (2002) first reported that cellulose can be dissolved without activation or pretreatment in, and regenerated from 1-butyl-3-methylimidazolium chloride ([Bmim][Cl]). Then a series of hydrophilic ionic liquids, such as 1-allyl-3-methylimidazolium chloride ([Amim][Cl]), 1-ethyl-3-methylimidazolium acetate ([Emim][OAc]), 1,5-diazabicyclo[4.3.0]non-5-enium acetate ([DBNH][OAc]), and so on, were proved to have outstanding capability for dissolving cellulose (Ren et al. 2003; Kosan et al. 2008; Parviainen et al. 2015). However, ionic liquids are not cheap enough for large-scale operations due to the high cost of synthesis. In view of this situation, deep eutectic solvents (DESs) as low cost alternatives to conventional ionic liquids have currently been a subject of extensive research, since their physico-chemical properties are close to those ionic liquids, which are constituted only by discrete ions, namely the range of conductivities, viscosities and temperature dependencies (Abbott et al. 2004a, b). Some authorities have classified DESs as a new generation of ionic liquids (Zhang et al. 2012a; Dai et al. 2013; Abbott et al. 2016).

At this point it seems as if DESs are capable to dissolve cellulose like ionic liquids. Unfortunately, to our best knowledge, only a few literature reports currently available (Lynam et al. 2017; Zdanowicz et al. 2018). Whether DESs are promising solvents for cellulose dissolution is still unclear. In this mini-review, we survey the current understanding of cellulose dissolved by DESs. The cellulose dissolution in DESs was discussed by summarizing the structure of cellulose and comparing with the dissolving mechanisms of NMMO and ionic liquids. Finally, a perspective on improving cellulose dissolution in DESs is offered.

Structure of cellulose related to dissolution

It is known that cellulose structure affects its affinity for dissolution. The primary structure of cellulose is very simple: a linear unbranched polymer of β-glucopyranoside residues, connected with $1 \rightarrow 4$ β-glycosidic bonds (Fig. 1). Two hydrogen bonds (between the C6 and the C2 hydroxyl and between the C5 oxygen and C3 hydroxyl) stabilize the glycosidic bond and make the structure stiff. The degree of polymerization of cellulose varies from 100 to 15,000 glucose units (Kongruang 2008).

The secondary structure of cellulose consists of alternating crystalline and amorphous regions. According to Ding's model (from maize parenchyma), the elementary fibril is synthesized from rosettes containing 36 cellulose synthase proteins that produce 36 β-D-glucans in higher plants. These 36 glucan chains then assemble through hydrogen bonds and van der Waals forces to form a crystalline core and the subcrystalline shell structure of the elementary fibril (Fig. 2). The hydrogen bonded supramolecular structure of cellulose makes it insoluble in most solvents and is partly responsible for the resistance of cellulose against microbial degradation (Gross et al. 2013; Jørgensen et al. 2007). Theoretically, cellulose dissolution is a process of hydrogen bond breaking, especially in the crystalline regions. The derivatizing solvents chemically react with hydroxyl groups and reduce the hydrogen bond network; the non-derivatizing solvents destroy inter- and intra-molecular hydrogen bonds either by hydrogen bond formation between one or more components of the solvent systems and the hydroxyl groups of the cellulose or by coordination bond formation between the metal ion present in the medium and the hydroxyl group of cellulose molecules (Trygg and Fardim 2011; Sen et al. 2013). The cellulose crystals can be easily investigated by the polarizing microscope. Therefore, polarizing microscopy is an important tool for judging cellulose dissolution.

Dissolution of cellulose in DESs

DESs were first reported by Abbott et al. (2001). They are systems formed from a eutectic mixture of Lewis or Brønsted acid and bases that contain a variety of anionic and/or cationic species. In contrast, conventional ionic liquids are formed from systems consisting of one type of discrete cation and anion (Smith et al. 2014). Although the concepts of DESs and ionic liquids are different, DESs are acknowledged as

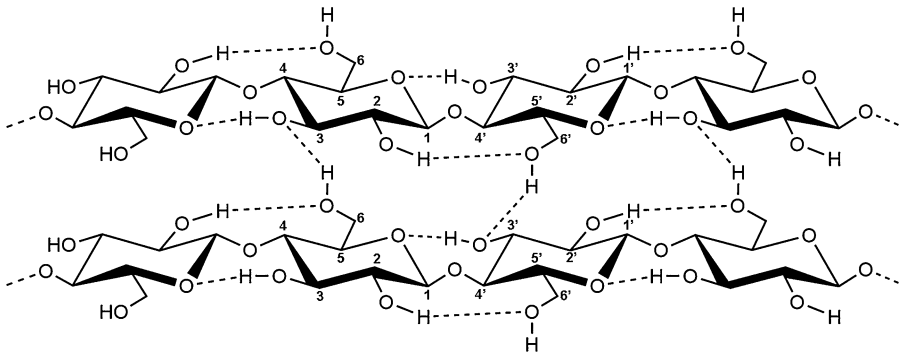


Fig. 1 Primary structure of cellulose and hydrogen bonds between cellulose chains. One hydrogen bond between C6 and C3 hydroxyls per glucose residue is formed between parallel chains (Ek 2009)

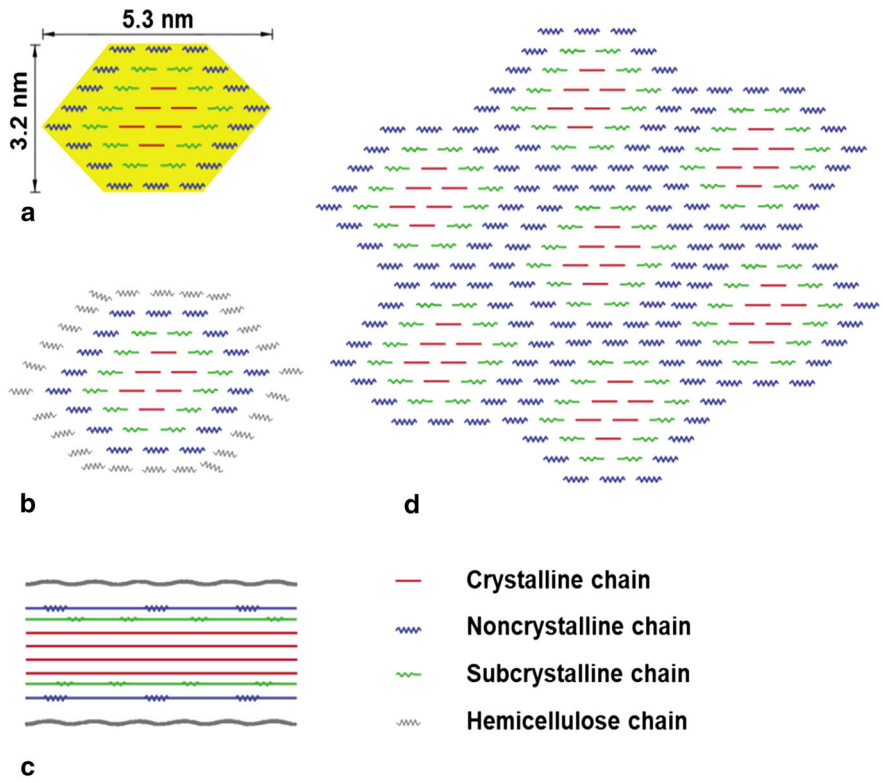


Fig. 2 Secondary structure of cellulose (Ding and Himmel 2006; Ding et al. 2012). **a** Cross-section of the elementary fibril containing 36 glucan chains. **b** Cross-section of the microfibril in which the hemicellulose is coated on the surface of the elementary fibril. **c** Longitudinal section of the microfibril. **d** Cross-section of the macrofibril formed by elementary fibrils

alternatives to ionic liquids as they share many characteristics and properties with ionic liquids.

In general, DESs are obtained by the complexation of a quaternary ammonium salt with a metal salt or hydrogen bond donor. Here, hydrogen bond donor refers to donation of the proton. Conversely, hydrogen bond acceptor refers to acceptance of the proton. One classic example is a mixture of choline chloride (ChCl) and urea in a 1:2 mol ratio (Abbott et al. 2003). The melting points of choline chloride and urea are 302 and 133 °C, respectively. The eutectic mixture can remain in the liquid state to temperatures as low as 12 °C due to the charge delocalization occurring through hydrogen bonding between halide ion and the hydrogen donor. DESs can be classified into four types depending on the nature of hydrogen bond acceptors and hydrogen bond donors used (Satlewal et al. 2018): (I) the non-hydrated metal halides and the quaternary ammonium salts (e.g. $ZnCl_2$ + choline chloride); (II) the hydrated metal halides and the quaternary ammonium salts (e.g. $CoCl_2{\cdot}6H_2O$ + choline chloride); (III) the hydrogen bond donor and the quaternary

ammonium salts (e.g. choline chloride + urea); (IV) the metal halides and the hydrogen bond donor (e.g. $ZnCl_2$ + urea). Structures of hydrogen bond acceptors and hydrogen bond donors used for formation of DESs are shown in Fig. 3. DESs are "designer solvents", which can change their physicochemical properties by a judicious selection of the constituents' nature and ratio.

Choline chloride is the most popularly employed member for synthesizing DESs as a hydrogen bond acceptor and thereby the studies on DESs for cellulose dissolution begun with choline chloride-derived DESs. Miller (2011) has tested the dissolution of various cellulosic polymers such as xanthan gum, modified guargum, carboxymethyl tamarind and sodium carboxymethyl cellulose in choline chloride urea eutectics. The cellulosic polymers were found to be soluble at 65 °C. Pan et al. (2017) found that amorphous cellulose in rice straw can be dissolved in choline chloride/urea, but the α-cellulose had inefficient solubility in choline chloride/urea. Even more unfortunately, microcrystalline cellulose was only sparingly soluble and almost no solubilization occurred in choline chloride/urea (molar ratio 1:2) (Tenhunen et al. 2017) and choline chloride/$ZnCl_2$ (molar ratio 1:2) (Zhang et al. 2012b). Hertel et al. (2012) further reported that no dissolution of MCC was observed in the DESs of choline chloride and other chemicals including oxalic acid, phenylacetic acid, malonic acid, glycerine, acetamide and formamide, indicating that the choline chloride derived DESs might be ineffective for dissolution of the original cellulose. In order to improve the solubility of cellulose in choline chloride derived DESs, Ren et al. (2016a) activated the cellulose by using ultrasound-assisted saturated calcium chloride solution. The

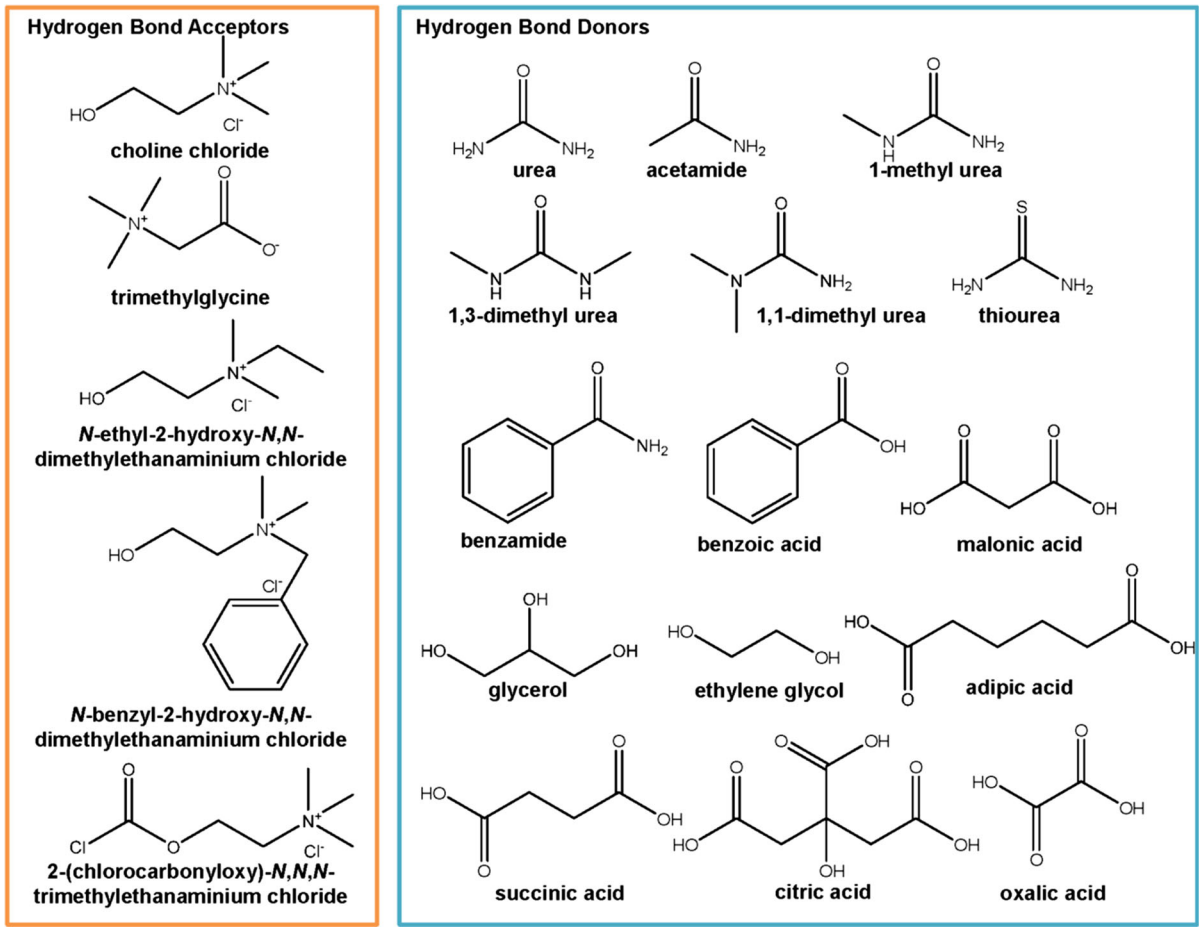


Fig. 3 Structures of some hydrogen bond acceptors and hydrogen bond donors used in formation of DESs (Smith et al. 2014)

solubility of cellulose was reported to be 1.43 wt% in choline chloride and urea. When the urea was replaced by imidazole, the novel DES had a higher solubility of cellulose (2.48 wt%). The solubility could be further improved to 4.57 wt% in choline chloride and imidazole by adding 5 wt% polyethylene glycol (PEG). Malaeke et al. (2018) then synthesized a series of DESs including choline chloride/phenol, choline chloride/α-naphthol, choline chloride/resorcinol and choline chloride/maleic acid. They reported that cellulose could dissolve in these DESs by using ultrasound irradiation. The maximum solubility of cellulose was found in choline chloride and resorcinol with 6.10 wt%.

To find other DESs for cellulose dissolution, Francisco et al. (2012) synthesized 21 kinds of DESs with different molar ratios. The maximum solubility of cellulose was reported to be 0.78 wt% in a DES comprising of malic acid and proline at 100 °C. Zhou and Liu (2014) designed a novel DES by using urea and caprolactam (molar ratio, 1:3). The maximum solubility of cotton-ramie pulp was 2.83 wt% at 50 °C. Despite the solubility of cellulose in these DESs was low, the results still gave us a hope to achieve the goal of higher cellulose solubility in DES. Ren et al. (2016b) designed a new allyl-functionalized choline DES to promote cellulose dissolution. The allyl-functionalized choline ([ATEAm]Cl) was first synthesized by using triethylamine and allyl chloride. Then the [ATEAm]Cl as a hydrogen bond acceptor and oxalic acid as a hydrogen bond donor were mixed and heated to obtain the product of [ATEAm]Cl^-Oxa. The maximum solubility of cellulose in this DES was reported to be 6.48 wt%. According to the literature, Table 1 presents the DESs that are currently reported to be a solvent for cellulose dissolution.

Perspective on improving cellulose solubility in DESs

Currently, the solubility of cellulose in DESs is far from that in NMMO and ionic liquids, which are two classic non-derivatizing solvents and have analogous physico-chemical properties to DESs. Based on the literature, the cellulose solution up to around 30 wt% can be produced in NMMO (Wang et al. 2016); the solubility of cellulose in ionic liquid [Emim][OAc] is more than 20 wt% (Sun et al. 2011). The reason of differences in solubility between DESs and these two solvents may be unveiled by reviewing the dissolving mechanisms of them.

As previously mentioned, cellulose dissolution is a process of hydrogen bond breaking. The solvents should have abilities to interrupt the inter- and intra-molecular hydrogen bonds of cellulose. In NMMO, swelling and dissolution of cellulose are caused by a far-reaching rearrangement of the hydrogen bond network in the system, meaning breakage of hydrogen bonding network in cellulose, with concomitant formation of new hydrogen bonds between solvent and cellulose (Adorjan et al. 2004). In ionic liquids, the anion and the cation play significant roles in the cellulose dissolution by forming hydrogen bonds between them and hydroxyl protons of cellulose (Wang et al. 2012; Lu et al. 2014). Similarly, the protic –OH groups in choline chloride derived DESs also have ability to react with cellulose (Ren et al. 2016b; Tenhunen et al. 2017). However, unlike NMMO or ionic liquids which are formed from a single polymer or a discrete anion and cation, the DESs are usually connected by their own hydrogen bonds (Abbott et al. 2004a). Hydrogen bond network in DESs results in less opportunities for interactions between solvent agents and cellulose. Accordingly, we suggest that competition to react with chloride between protic –OH groups in the choline cation and hydrogen bonds in the cellulose causes decreased solubility, and therefore the permeability of DESs for cellulose is limited. Such a situation won't occur in NMMO and ionic liquids since there are no abundant hydrogen bonds formed in the solvents (Rosenau et al. 2001; Werner et al. 2010). Therefore, the low solubility of cellulose in DESs is unsurprising. The DESs are not ideal solvents for cellulose dissolution at present.

Theoretically, if we could design a novel DES to create more opportunities for interactions between solvent agents and cellulose and enhance their permeability, the solubility of cellulose in DESs would be improved. To achieve this goal, the hydrogen bond accepting ability of DESs should be increased at first. Previous studies on designing ionic liquids for dissolving cellulose have already solved a similar question (Itoh 2014). In ionic liquids, the solubility of cellulose increases almost linearly with the increasing hydrogen bond accepting ability of the anions and cations (Ohno and Fukaya 2009). The Kamlet–Abboud–Taft parameter (KAT values), which

Table 1 Reported DESs for cellulose dissolution

Raw material	DESs	Molar ratio	Method	Cellulose solubility (wt%)	References
Microcrystalline cellulose AVICEL PH 105	ChCl/urea	1:2	Processing at 110 °C	< 0.2	Zhang et al. (2012b)
Same as above	ChCl/$ZnCl_2$	1:2	Same as above	< 0.2	Same as above
Cellulose (≥ 90% mass fraction purity) purchased from Sigma-Aldrich)	Malic acid/alanine	1:1	Processing at 100 °C	0.11	Francisco et al. (2012)
Same as above	Malic acid/glycine	1:1	Same as above	0.14	Same as above
Same as above	Malic acid/proline	1:2	Same as above	0.24	Same as above
Same as above	Oxalic acid dihydrate/histidine	9:1	Processing at 60 °C	0.25	Same as above
Same as above	Malic acid/proline	1:3	Processing at 100 °C	0.78	Same as above
Cotton-ramie pulp (DP = 517)	Acetamide/urea	2:1	Processing at 50 °C	1.03	Zhou and Liu (2014)
Same as above	Acetamide/caprodacine	1:1	Same as above	1.79	Same as above
Same as above	Caprolactam/urea	3:1	Same as above	2.83	Same as above
Cotton linter pulp (DP = 575.6)	ChCl/caprolactam	1:1	Activating cellulose by ultrasound-assisted saturated calcium chloride solution prior dissolution; Processing at 120 °C	0.16	Same as above
Same as above	ChCl/acetamide	1:2	Same as above	0.22	Ren et al. (2016a)
Same as above	ChCl/ammonium thiocyanate	1:1	Same as above	0.85	Same as above
Same as above	ChCl/urea	1:2	Same as above	1.43	Same as above
Same as above	ChCl/imidazole	3:7	Same as above	2.48	Same as above
Cotton linter pulp (DP = 575.6)	Triethyl-allyl ammonia chloride/oxalic acid	1:1	Processing at 110 °C	6.48	Ren et al. (2016b)
Cellulose (≥ 98% mass fraction purity) purchased from Merck	ChCl/maleic acid	1:1	Processing with ultrasonic irradiation (20 Hz) at 90 °C	2.57	Malaeke et al. (2018)
Same as above	ChCl/α-naphthol	1:1	Same as above	3.39	Same as above
Same as above	ChCl/phenol	2:1	Same as above	4.70	Same as above
Same as above	ChCl/resorcinol	1:1	Same as above	6.10	Same as above

specifies three distinct solvent polarities including hydrogen bonding acidity (α), hydrogen bonding basicity (β), and dipolarity/polarizability (π*), is a useful tool for estimating such ability of solvent (Kamlet and Taft 1976). A general procedure for determination of KAT values was given by Eyckens et al. (2016). Results show that the stronger the hydrogen bond acceptance of the ionic liquid, the stronger the ability of the ionic liquid to dissolve cellulose. In this case, the β-values obtained in a solvate chromatic study might be a good indicator of the ability to dissolve cellulose since high hydrogen bonding basicity can weakens the inter- and intra-molecular hydrogen bonds in cellulose crystal (Abe et al. 2010). This theory may also be applicable to the DESs for cellulose dissolution. Here, we propose a strategy that the components which have anions of high hydrogen bonding basicity such as Cl^-, OAc^-, $HCOO^-$, $(MeO)_2PO_2^-$ are promising choices to form DESs for cellulose dissolution as well as the special chemical structures of cations in ionic liquids shown in Fig. 4, such as morpholine and imidazole. In addition, other assistant methods should be taken into consideration to enhance the permeability of DESs.

Our strategy is exemplified by the data in Table 1. The imidazole and the allyl group are special chemical structures with good hydrogen bond accepting ability. Therefore, it is found that the maximum solubility of cellulose in choline chloride and imidazole (2.48 wt%) is higher than that in choline chloride and urea(1.43 wt%); the allyl-functionalized choline DES reaches the highest solubility of cellulose among the listed DESs (6.48 wt%). The data in Table 1 also show that the original cellulose is difficult to dissolve in DES comprising choline chloride and urea (< 0.2 wt%) while the solubility is increased by activating the cellulose with ultrasound irradiation. We suggest that the function of ultrasound irradiation in this case is similar to that in ionic liquids: the ultrasound irradiation provides a greater penetration of solvents into cellulose and improves mass transfer (Lan et al. 2011). Consequently, ultrasound-assisted treatment may be an effective way to improve cellulose solubility by enhancing the permeability of DESs.

It should be noted that our strategy is based on an underlying hypothesis that the cellulose insolubility is due to the strong intermolecular hydrogen bonds between cellulose molecules and that bringing cellulose into solution is dependent on breaking these

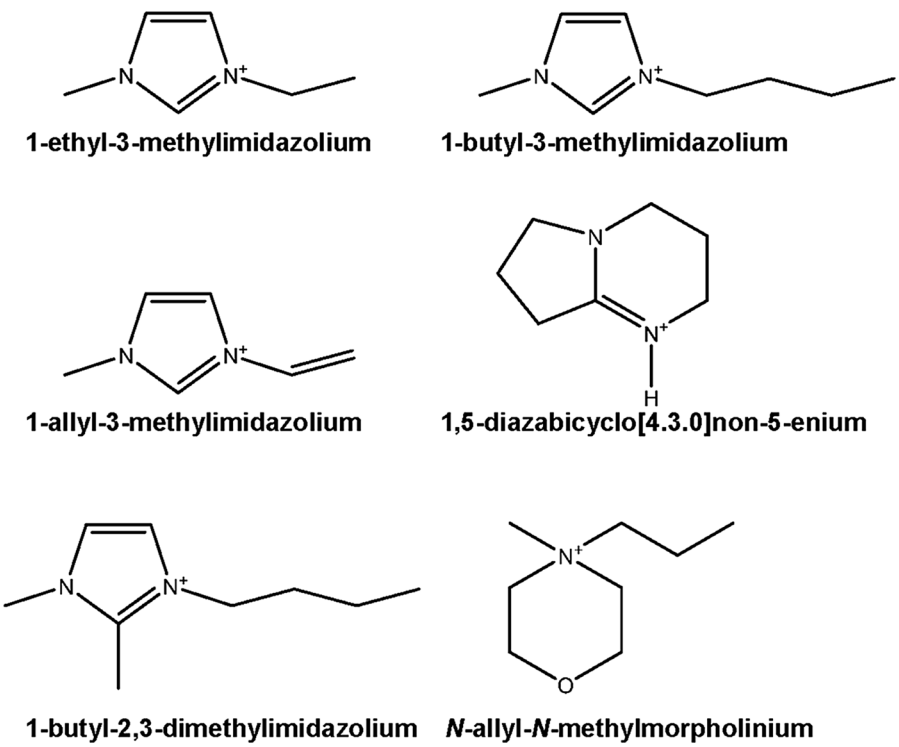


Fig. 4 The special chemical structures of cations for designing novel DESs

hydrogen bonds. Lindman et al. (2010) suggested that this hydrogen bonding theory cannot explain the low aqueous solubility of cellulose. According to their evidence, the low aqueous solubility must have a marked contribution from hydrophobic interactions and co-solutes, which have the tendency to weaken hydrophobic interactions, would be expected to promote aqueous solubility of cellulose. Examples of such compounds are urea and its derivatives as well as PEG. As previously mentioned, because DESs are also strongly hydrogen-bonded liquid like water, this theory may be applicable to promote the cellulose solubility in DESs and can explain why the PEG can promote the cellulose solubility in DES comprising choline chloride and imidazole (Ren et al. 2016a). In this case, PEG served as a surfactant to reduce hydrophobicity of cellulose and thereby enhanced the permeability of DES.

Except for improvement of cellulose dissolution in DESs, we notice that cellulose also can be regarded as a hydrogen bond donor, indicating that cellulose has potential to form a DES. If this idea works, a new and promising opportunity may be opened up to process cellulose in DESs (Tang et al. 2017). However, no one has yet produced a detailed case based on the use of cellulose and other chemicals to support or reject this hypothesis. A great deal of work remains to be done to achieve this goal. In any case, DESs are promising solvents for cellulose dissolution. More innovation and deeper investigations are definitely needed. Ionic liquids were first reported in 1914, but we didn't know they are good solvents for cellulose dissolution until the year 2002 (Swatloski et al. 2002). We should never stop looking for new DESs that are capable to dissolve cellulose.

Acknowledgments We thank the National Key R&D Program of China (2017YFD0601004) for financial support.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

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